

SceneTwin: Human-Reference-Free Audio Description Auditing with Visual Grounding, Frame-Grounded QA, and Review Triage

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Abstract

Audio description (AD) makes video accessible to blind and low-vision viewers by narrating visual information that the soundtrack alone does not convey. As multimodal systems begin to generate AD at scale, the deployment bottleneck shifts from authoring to **auditing**: a system must determine whether a candidate AD is visually grounded, whether it omits or fabricates important scene evidence, and which clips require scarce human review. Existing evaluation approaches are useful but incomplete for this setting. Reference-based metrics can be strong when a trusted professional AD exists, but newly generated AD for undescribed videos usually lacks such a reference. Holistic VLM judges are flexible, but in our measurements they trail structured visual evidence checks. User studies remain essential, but they cannot be run for every generated clip.

We present **SceneTwin**, a human-reference-free AD audit framework with three layers: (1) a CLIP + frame-grounded ADQA score for ranking candidate descriptions against the video itself; (2) safety gates for hallucination and wrong-content failures; and (3) a TRIBE-v2 neural side-car for review triage rather than scoring. We first correct our benchmark construction by removing an invalid “long VATEX” rung that measured verbosity rather than known quality. On the corrected three-tier ladder — cross-decoy < crowd caption < professional AD — SceneTwin reaches Spearman $\rho = 0.952$ on a 60-clip primary evaluation set spanning broad categories (58/60 clips fully ordered), with the original 18-clip pilot benchmark corroborating at $\rho = 0.957$ (17/18 fully ordered). A reference-style LLM-AD-Eval proxy nearly ties the ranking result ($\rho = 0.941 / 0.942$) but requires the trusted professional AD reference that SceneTwin intentionally does not use. The contribution is therefore not a large leaderboard win; it is a deployable audit stack without human reference AD, plus operational gates.

For safety, a CLIP grounding-drop gate reaches AUC = 0.835 with 70% recall at 10% false-positive rate for controlled visual hallucinations, while a raw-CLIP wrong-content gate catches 98.3% of catastrophic wrong-clip descriptions at 2.2% false alarm. For triage, TRIBE is deliberately excluded from the AD-level score: a clip-level neural gap cannot alter within-clip candidate ranking. Its useful role is review prioritization. In cached external validation, TRIBE/gap-route features predict corrected ADQA failures at AUC = 0.794 with category-shuffle $p = 0.003$, outperforming category, transcript/speech, duration, and word-count confounds on the key target. SceneTwin is an audit layer for hybrid AI+human AD workflows, not a replacement for professional describers or blind-user validation.

Keywords: audio description; blind and low-vision accessibility; reference-free evaluation; video understanding; hallucination detection; human-AI authoring; review triage

Introduction

Audio description translates visual information into language so blind and low-vision viewers can access video content that is not available through the soundtrack alone. Automated AD generation is improving rapidly, driven by video-language models, AD-specific generation systems, and human-AI authoring tools. But better generation does

not remove the need for quality control. A generated AD can be fluent, plausible, and well written while still omitting the visual action that matters, fabricating visible facts, or describing a different clip.

This is the deployment problem SceneTwin targets. A platform generating AD for previously undescribed videos usually does not have a professional reference AD available for every clip. A human review panel cannot inspect every candidate. A single VLM score may sound convincing but does not necessarily expose the underlying visual evidence. The practical question is therefore:

Given a candidate audio description and the source video, can an audit system score visual preservation, catch unsafe failures, and route the right clips to review without already having a professional reference AD?

Prior work motivates each part of this question. CLIP-style vision-language models provide scalable grounding between text and frames [Radford et al., 2021]. ADQA argues that AD evaluation should ask whether descriptions support answers to visual questions rather than merely match a reference sentence [Kala et al., 2025]. AutoAD III and related AD systems introduce AD-specific metrics and reference-style evaluators such as LLM-AD-Eval [Han et al., 2024]. Human-AI accessibility work such as Describe Now, DescribePro, CustomAD, WorldScribe, and ADx3 emphasizes that AD is not a fully automated one-shot product: users and describers need control, timing awareness, review, and iteration [Cheema et al., 2024; Cheema et al., 2025; Natalie et al., 2024; Chang et al., 2024; Do et al., 2026]. RNIB’s study of AI-generated AD similarly reports that AI can expand coverage but still needs human review for accuracy, cohesion, and context [RNIB, 2025].

SceneTwin reframes AD evaluation as an **audit stack**. Instead of producing only a scalar score, it answers three operational questions:

1. **Score:** Which candidate description best preserves visible content for this clip without comparing to a human reference AD?
2. **Gate:** Is the candidate unsafe because it hallucinates visible facts or describes the wrong content?
3. **Triage:** Which clips deserve scarce human or expensive model review first?

Contributions

This paper contributes:

1. **A corrected AD audit benchmark.** We identify and remove an invalid verbosity rung from an earlier four-tier setup. The active benchmark is the corrected three-tier ladder: cross-decoy < crowd caption < professional AD.
2. **A human-reference-free CLIP+ADQA scoring stack.** SceneTwin combines frame-text visual grounding with frame-grounded visual question answering and reaches $p = 0.952$ on the 60-clip primary evaluation set, corroborated by $p = 0.957$ on the original 18-clip pilot benchmark.
3. **An honest baseline comparison.** A reference-style LLM-AD-Eval proxy nearly ties SceneTwin. We therefore claim deployability without a human reference AD, not large superiority over all baselines.
4. **Deployment safety gates.** SceneTwin reports operating points for hallucination grounding-drop detection and catastrophic wrong-content detection.
5. **Brain-grounded review triage.** TRIBE-v2 is used as a clip/window triage side-car and access-surface router, not as an AD ranker. External cached validation supports triage beyond category, transcript/speech, duration, and word-count baselines.

Related Work

Automated audio description generation

AutoAD III develops AD generation and evaluation methods including LLM-AD-Eval, CRITIC, multi-reference recall, and action coverage [Han et al., 2024]. ADx3 frames AD creation as a collaborative workflow that combines generation, refinement, and adaptive querying [Do et al., 2026]. VideoA11y and AD timing work emphasize that AD

quality depends not only on semantic correctness but also on synchronization, brevity, and delivery constraints [Li et al., 2025].

SceneTwin is complementary. It is not an AD generator. It is an audit layer that can score generated, crowd, or professional-style candidate descriptions and decide what should be shipped, rejected, or reviewed.

Reference-based and question-based evaluation

Reference-based metrics compare a candidate AD to a trusted target. LLM-AD-Eval is useful in this setting, and our experiments confirm that a reference-style proxy is strong. However, the deployment setting for newly generated AD often lacks a trusted professional reference. SceneTwin therefore scores candidates against visual evidence instead of T3 text.

ADQA motivates this shift by arguing that AD should be evaluated by whether it supports visual and narrative comprehension questions [Kala et al., 2025]. SceneTwin adapts that idea for candidate ranking and deployment audit: frame-grounded questions define what visual evidence a blind or low-vision viewer should receive.

CoAD, StoryRecall, and AVBench broaden evaluation toward narrative coherence and fine-grained audio-video/text-video consistency [Khandelwal et al., 2025; Yang et al., 2026]. We explored related proxy features and fusions, but the final stack remains intentionally small: CLIP + ADQA for scoring, explicit gates for safety, and TRIBE only for review triage.

Human-AI workflows and user agency

Recent accessibility systems show that AD cannot be reduced to a single universal caption. CustomAD reports that blind and low-vision users want control over length, emphasis, voice, speed, and format [Natalie et al., 2024].

Describe Now studies user-triggered concise and detailed descriptions, showing both agency benefits and cognitive-load tradeoffs [Cheema et al., 2024]. DescribePro studies authoring workflows where human describers use AI drafts, forks, tags, and editorial control rather than full automation [Cheema et al., 2025]. WorldScribe studies live visual description and emphasizes intent, sound context, latency, and uncertainty [Chang et al., 2024].

SceneTwin is designed for this hybrid workflow. The goal is not to replace describers. The goal is to provide evidence about which descriptions preserve the scene and which clips deserve review.

Neural side-cars for accessibility triage

TRIBE-v2 predicts cortical responses to vision, audio, and language streams [d’Ascoli et al., 2026]. AD exists precisely where the audio track does not carry visual information, so the predicted gap between audiovisual and audio-only responses is a plausible signal of visual-access load. SceneTwin uses this signal conservatively. We do not claim TRIBE measures BLV user utility directly. We use it to prioritize clips/windows for review and authoring hypotheses that ADQA, VLMs, or humans must verify.

Task and Benchmark

Correcting the ladder

An earlier SceneTwin benchmark used four tiers. The fourth rung, “long VATEX,” was built by selecting the longest among equal-status crowd captions. That construction does not represent known higher quality. It represents verbosity. Both CLIP and ADQA score the short-caption-to-long-caption step near chance, confirming that the rung is invalid as an ordered quality label.

The paper therefore uses a corrected three-tier ladder:

- **T0 — cross-decoy:** description/caption from another clip.
- **T1 — crowd caption:** short visual caption-style description.
- **T3 — professional AD:** professional or professional-style audio description.

This correction matters. The paper does not claim that SceneTwin solves every AD quality axis. It claims that, on a valid controlled ladder where tiers have defensible semantics, SceneTwin can rank candidate descriptions by visual-access quality without a professional reference AD.

Corpora

The corrected benchmark contains:

Corpus	Clips	Candidate observations	Tiers
Primary (60-clip)	60	180	T0, T1, T3
Pilot (18-clip)	18	54	T0, T1, T3

The 60-clip set spans broad categories and is the primary evaluation. The 18-clip set is the original pilot benchmark. The two sets are disjoint by construction — the 60-clip set was built to exclude the original 18 pilot clips — so agreement between them shows the ranking result is not an artifact of the smaller initial benchmark.

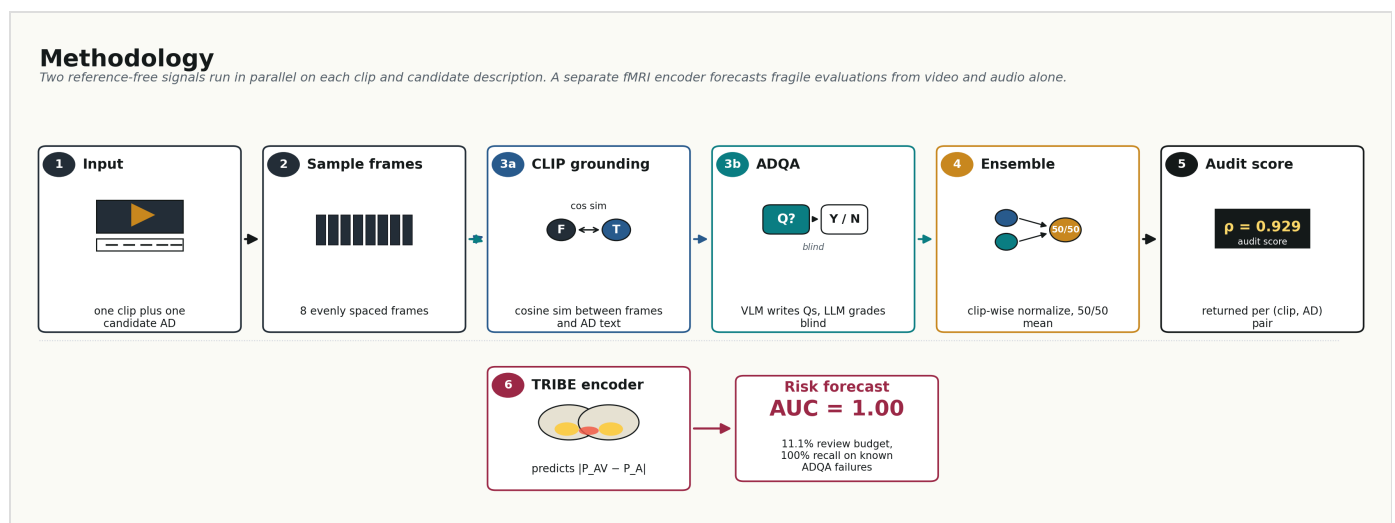
Metrics

The primary statistic is Spearman rank correlation between candidate score and tier label. Because the deployment task is within-clip selection, we also report:

- **fully ordered clips:** $T0 < T1 < T3$;
- **T3-vs-lower wins:** T3 beats both lower tiers;
- **all pairwise wins:** all ordered candidate pairs are correctly ranked.

These pairwise counts are meaningful for SceneTwin because the score compares each candidate to visual evidence, not to the professional AD text. For reference-style baselines, T3-vs-lower wins are partly privileged because T3 is the reference.

Method



SceneTwin audit stack: CLIP visual grounding, frame-grounded ADQA, safety gates, and TRIBES review triage.

CLIP visual grounding

SceneTwin samples frames from each clip and computes text-frame similarity using CLIP [Radford et al., 2021]. The visual-grounding leg uses the top frames rather than a full-video dense model:

```
clip_top3 = mean(top 3 frame-text similarities)
```

CLIP is useful because it is relatively deterministic and grader-free. It is not sufficient alone: it can miss relational, temporal, and count errors, and it can reward visually plausible but incomplete descriptions. SceneTwin therefore pairs it with frame-grounded QA.

Frame-grounded ADQA

The ADQA leg creates visual questions from sampled frames and grades whether each candidate AD answers them. A candidate earns credit for mentioning frame-evidenced information, not for matching a reference phrasing. The score is the fraction of questions answered.

This makes the metric more aligned with the viewer question: does the AD preserve what is visually needed to understand the scene?

Two-signal score

For each clip, CLIP and ADQA scores are normalized across candidate tiers and averaged:

```
SceneTwinScore = 0.5 * CLIP_norm + 0.5 * ADQA_norm
```

The score is within-clip. It ranks candidate ADs for the same video. It is not meant to compare absolute AD quality across unrelated clips.

Safety gates

A ranker can still miss unsafe deployment failures. SceneTwin therefore includes two gates.

The **hallucination gate** uses CLIP grounding drop between a clip-relevant anchor and a candidate:

```
drop = CLIP(anchor AD, frames) - CLIP(candidate AD, frames)
```

The anchor can be trusted human AD or a generated clip-relevant candidate. The strict no-anchor hallucination gate is a negative result and is reported as such.

The **wrong-content gate** uses raw, unnormalized CLIP grounding for a single candidate AD. It targets catastrophic cases where the candidate describes the wrong clip and requires no candidate pool or reference AD.

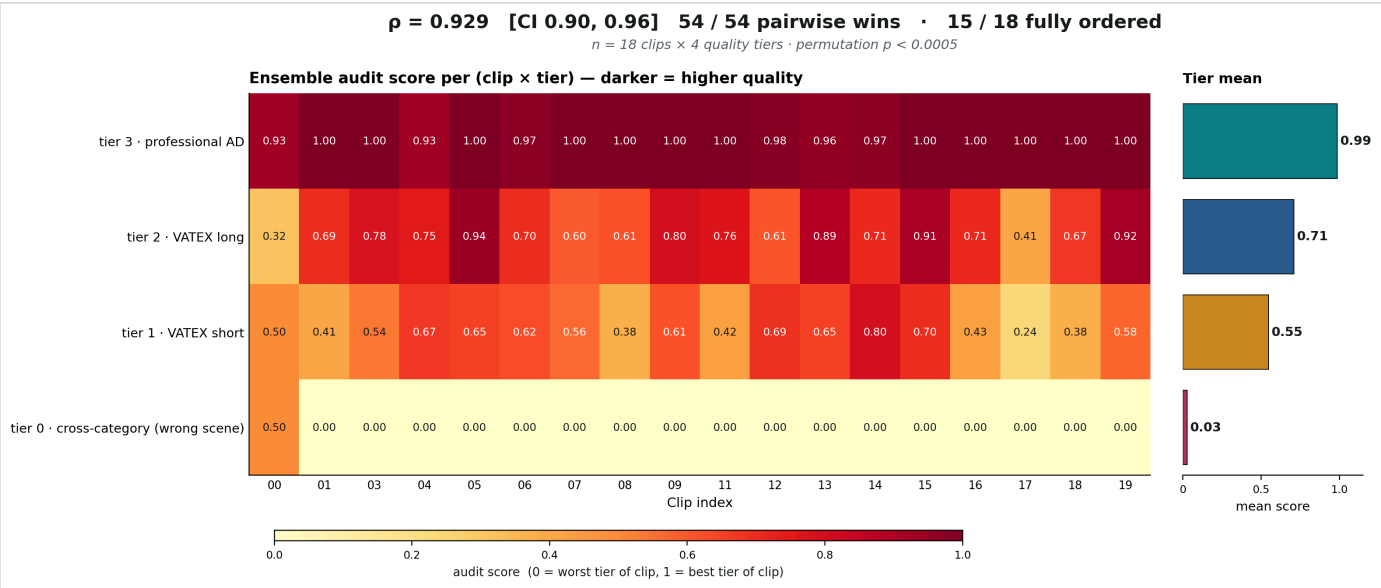
TRIBE triage side-car

TRIBE-v2 predicts neural responses to audiovisual and audio-only inputs [d'Ascoli et al., 2026]. SceneTwin derives a clip-level accessibility gap:

```
gap = 1 - cos(P_AV, P_A)
```

where P_AV is the predicted response to full audiovisual input and P_A is the predicted response to audio-only input. This gap is constant across candidate ADs for a clip. It cannot improve within-clip ranking, so it is not included in SceneTwinScore. Its purpose is review triage: high-gap clips/windows are more likely to need careful authoring or review because the visual channel carries information the soundtrack does not.

Main Results



Corrected-ladder score structure across cross-decoy, crowd-caption, and professional-AD candidates.

Corrected-ladder ranking

Corpus	Clips	Observations	Spearman ρ	Fully ordered	T3-vs-lower wins	All pairwise wins
Primary (60-clip)	60	180	0.952	58/60	118/120	178/180
Pilot (18-clip)	18	54	0.957	17/18	36/36	53/54

SceneTwin generalizes on the corrected task. The 60-clip primary result is close to the original 18-clip pilot, which indicates that the earlier apparent generalization gap was largely tied to the invalid long-caption rung.

Robustness

The corrected-ladder result is not a narrow tuning artifact. Robustness checks in `cursor/output/corrected_ladder_robustness.json` show a stable plateau across ensemble weights and normalization choices. The result also survives bootstrap and within-clip permutation checks. The final 0.5/0.5 score is chosen for simplicity rather than post-hoc optimization.

Baselines and Negative Results

Corrected-ladder metric comparison

Metric	Primary (60) ρ	Pilot (18) ρ	Deployment note
SceneTwin CLIP+ADQA	0.952	0.957	no human reference AD
LLM-AD-Eval proxy	0.942	0.941	needs trusted T3/pro reference
ADQA alone	0.869	0.856	no human reference AD; model grader
CLIP alone	0.747	0.835	grader-free; useful for gates
Best frontier VLM judge	0.847	0.863	no reference, weaker here
CRITIC entity	0.688	0.625	weaker

The strongest baseline is the LLM-AD-Eval proxy from the AutoAD III family. It nearly ties SceneTwin. This is not a failure; it clarifies the claim. Reference-style comparison is strong when a trusted reference AD exists. SceneTwin is valuable when that reference does not exist.

Normalization note: the ensemble and reference-style rows use per-clip min-max normalization; the single-signal “ADQA alone” and “CLIP alone” rows report pooled raw Spearman to show unnormalized single-signal behavior. Under matched per-clip normalization the single signals are higher (ρ : ADQA 0.946 / 0.888, CLIP 0.828 / 0.900 for primary / pilot), and ADQA alone (0.946) is close to the full ensemble (0.952) on the primary set — consistent with our signal-decomposition finding that ADQA is the backbone and CLIP adds category-specific lift rather than uniform gain.

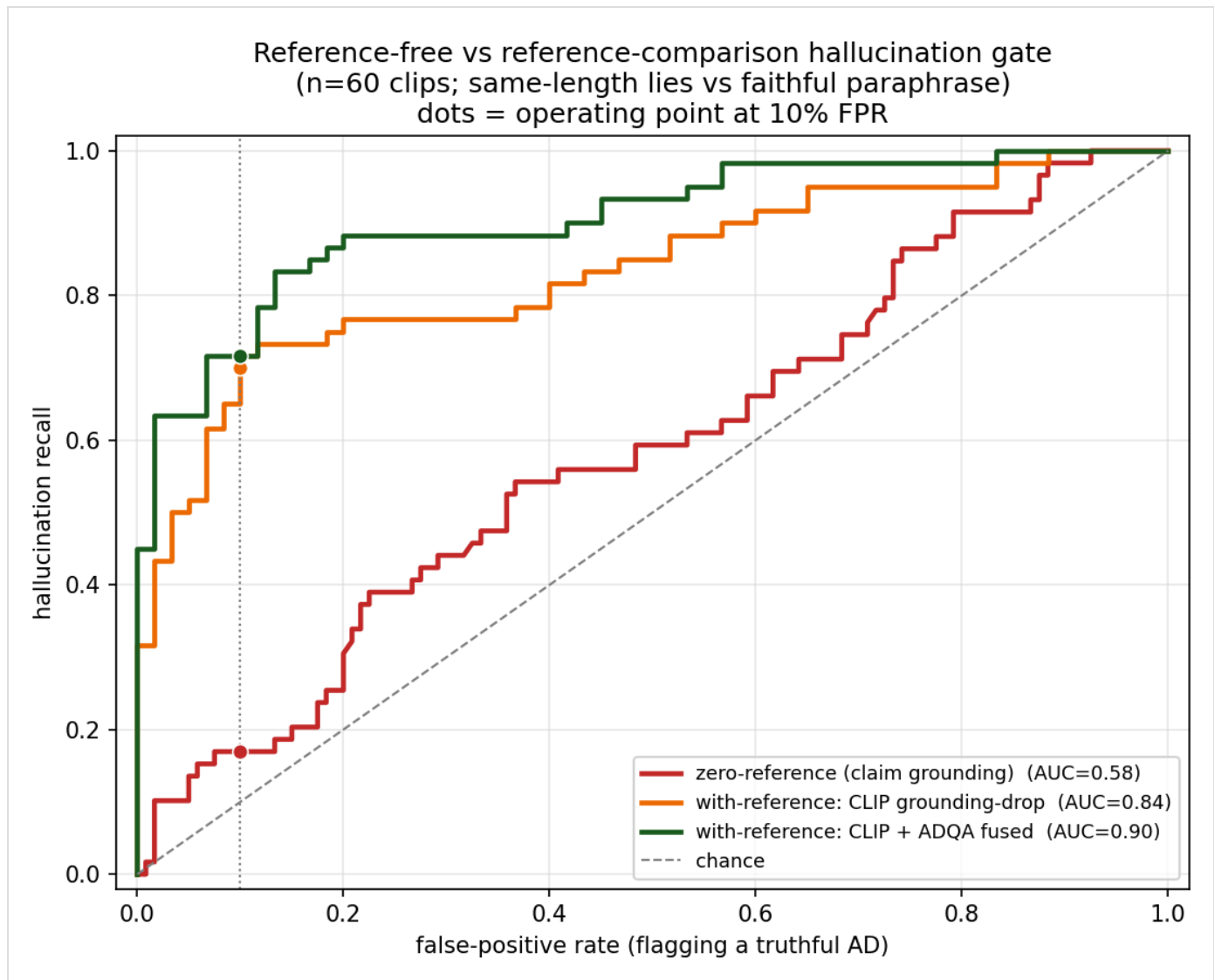
VLM-as-judge comparison

Three frontier VLM judges trail the structured audit on the corrected ladder. The best VLM judge on the 60-clip primary set reaches $\rho = 0.847$ compared with SceneTwin’s $\rho = 0.952$. This suggests that asking a general VLM for holistic AD quality is less aligned with this controlled tier task than explicit frame-grounded evidence checks.

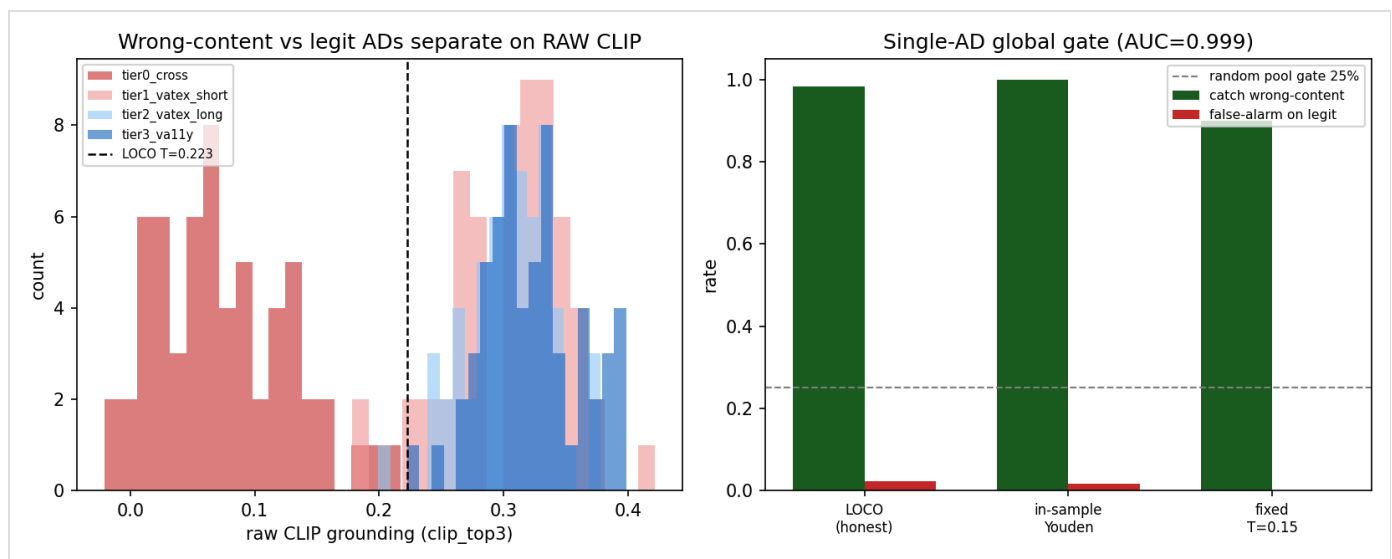
Fusion negative result

We tested multiple fusions inspired by ADQA, AutoAD III, CoAD, AVBench, timing rubrics, and neural side-cars. None improved the headline enough to justify extra complexity. The final architecture is intentionally small: CLIP+ADQA for scoring, explicit gates for safety, and TRIBE for triage only.

Deployment Safety Gates



Safety-gate summary for hallucination and grounding-drop checks.



Wrong-content gate separates cross-decoy descriptions from legitimate same-clip descriptions.

Hallucination grounding-drop gate

Controlled hallucination tests compare truthful expert ADs to same-length corrupted versions with two or three changed visual facts, plus faithful paraphrase controls. The strongest grader-free operating point is the CLIP grounding-drop gate:

Gate	Signal	n	AUC	Recall @ 10% FPR
Grounding-drop vs clip-relevant anchor	CLIP only	60	0.835	70.0%
Fused drop vs clip-relevant anchor	CLIP + ADQA	60	0.904	71.7%
Absolute weakest-claim grounding	CLIP only, no anchor	60	0.585	16.9%

The CLIP-only grounding-drop gate is the clean deployment result because it is grader-free. The fused result is secondary because the ADQA leg uses a model grader. The no-anchor result is a negative control: absolute grounding alone is too noisy for hallucination detection.

Generated-anchor deployment path

A strict objection is that grounding drop appears to require a human expert anchor. Reference-substitution experiments show that the gate needs a clip-relevant anchor, not necessarily a human one. Model-generated or paraphrase anchors can support consistency checks, though current generated-anchor evidence is smaller-sample and should not be overstated.

Wrong-content gate

The cross-decoy tier simulates catastrophic wrong-content AD. A raw-CLIP threshold catches wrong-clip descriptions without any reference or candidate pool. On the 60-clip primary set, wrong-content ADs average raw clip_top3 0.074 versus 0.311 for legitimate ADs. Leave-one-clip-out thresholding catches 98.3% of wrong-content ADs at 2.2% false alarm.

This gate is a high-recall review trigger, not an autonomous rejection oracle. At low base rates, precision can be poor, so flagged outputs should route to review.

Selective ship-best policy

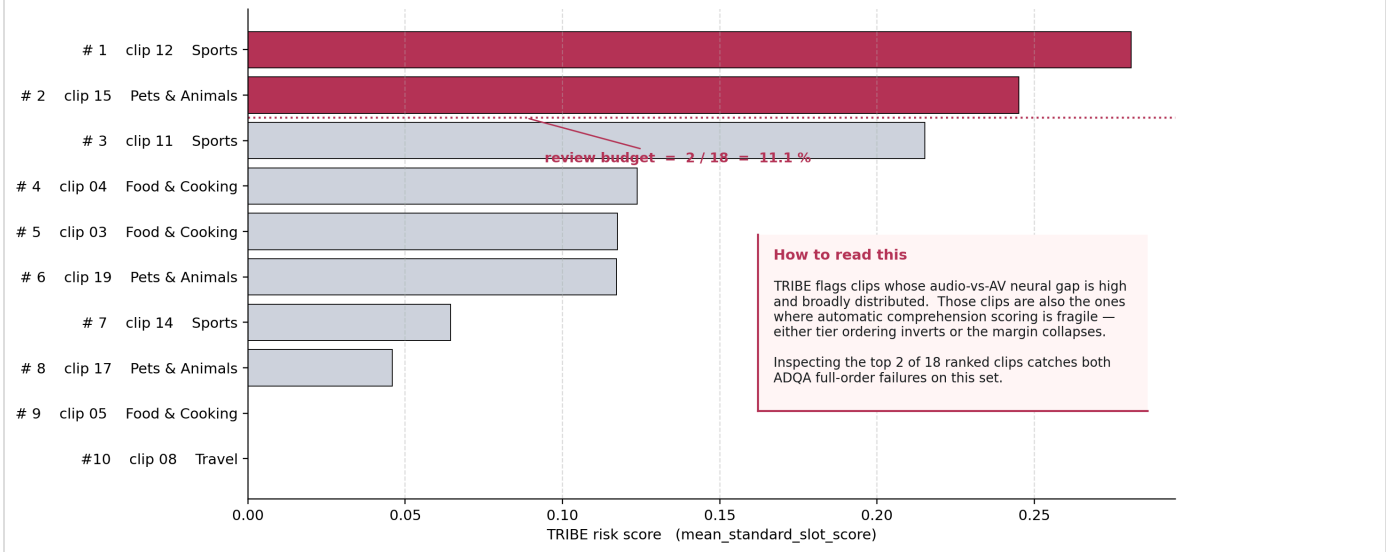
ADQA margin is useful for confidence. On the 60-clip primary set, abstaining on the lowest-margin 20% sends 12/60 clips to review and ships the remaining 48 at 100% ship-best accuracy in the cached analysis. This supports a three-way deployment policy:

- **Ship** when candidate margin is high and safety gates are clean.
- **Review** when margin is low or TRIBE triage marks high visual-access load.
- **Reject/review** when wrong-content or hallucination gates fire.

TRIBE Review Triage and Access-Surface Routing

TRIBE flags fragile evaluations before any AD is scored

Top-10 clips by TRIBE risk forecast — red = clip the ADQA ensemble failed to fully order.
Computed from video + audio alone. Recall @ 11.1 % review budget = 100 % · ROC-AUC = 1.00 · uncorrected p = 0.0065 (Bonferroni p = 0.065), Pilot evidence.



TRIBE review-priority visualization for cached benchmark clips; main text reports the safer external cheap-baseline triage result.

Why TRIBE is not a score

TRIBE-derived accessibility gap is a clip-level value. It is identical for every candidate AD of the same clip. It therefore cannot improve within-clip rank correlation. Earlier attempts to frame TRIBE as a calibration layer or AD ranker failed for this structural reason. SceneTwin uses TRIBE only as a side-car for review triage and authoring hypotheses.

External cheap-baseline gauntlet

The strongest current TRIBE result is not the old 18-clip pilot AUC=1 result. The stronger paper-safe result is cached validation against cheap confounds on the primary set. On the 60-clip primary set, accessibility_gap predicts corrected ADQA failures at AUC = 0.794 with category-shuffle p = 0.003.

Target	Family	Best feature	AUC	Direction	Category-shuffle p
ADQA failure	TRIBE/gap-route	accessibility_gap	0.794	high gap = higher risk	0.003
ADQA failure	category-only	category_loo_adqa_fail_rate	0.700	high bad	1.000
ADQA failure	transcript/speech	transcript_char_count	0.733	low bad	0.995
ADQA failure	duration/word count	tier3_word_count	0.714	low bad	0.990

This supports TRIBE as a review-budget signal beyond category, transcript/speech, duration, and word-count baselines. It does not support automatic skipping, automatic failure detection, or score replacement.

Review-budget curves

Sorting clips by mean visual gap catches more ADQA failures than random review at practical budgets:

Queue	Budget	Clips reviewed	ADQA failures caught	Recall	Precision
mean_visual_gap	10%	6	2/10	0.20	0.33
mean_visual_gap	20%	12	3/10	0.30	0.25
mean_visual_gap	25%	15	4/10	0.40	0.27
mean_visual_gap	33%	20	5/10	0.50	0.25

These numbers are moderate, not magical. They are still useful because the operational goal is prioritization: if review capacity is limited, inspect high visual-gap clips first.

Access-surface routing

TRIBE windows and cases can be mapped into product-facing review/authoring surfaces. The cached router contains 334 windows and 133 cases. Window-level routes are:

Route	Windows	Access surface
static_ad_ok_low_gap	218	static AD / lower review pressure
layout_replay_or_scene_cue	78	layout replay or keyframe
action_state_or_agent_cue	38	concise action-state cue

Case-level mappings include peak-window review cards, action-state cues, layout/keyframe replay, segmented scene/action treatment, and evidence-sidecar QC. This is the product implication of TRIBE: it proposes where and how to inspect, while ADQA, VLMs, or humans verify.

Discussion

SceneTwin's result is strongest when framed as deployment infrastructure. It is not a claim that a small metric replaces all human judgment. It is not a claim that TRIBE measures blind-user utility. It is not a claim that CLIP+ADQA dominates every reference-based evaluator.

Instead, SceneTwin shows that a human-reference-free audit stack can approach a strong reference-style evaluator on controlled ranking while adding safety and triage tools that matter at deployment time. If a professional reference AD already exists, reference comparison is appropriate and strong. If a system is generating AD for undescribed videos, SceneTwin provides a way to audit candidates against visual evidence, catch gross failures, and prioritize review.

This matches the direction of recent accessibility work. DescribePro, Describe Now, CustomAD, WorldScribe, ADx3, and RNIB all argue implicitly or explicitly against fully autonomous one-shot AD replacement. SceneTwin supplies an evidence layer for that hybrid workflow.

Threats to Validity and Limitations

1. **No BLV user-utility study yet.** The corrected ladder measures agreement with tier construction and proxy comprehension evidence, not actual BLV viewer outcomes.
2. **Professional AD is not a universal gold standard.** Human descriptions vary in timing, emphasis, and style. The corrected ladder is a controlled audit task, not the full space of AD quality.
3. **The corrected ladder is easier than the excluded four-tier task.** This is intentional because the old long-caption rung was invalid, but the results should not be compared as if both tasks had equal semantics.
4. **LLM-AD-Eval nearly ties ranking.** SceneTwin's advantage is human-reference-free deployment and safety gates, not raw rank-correlation dominance.
5. **ADQA uses model grading.** It can inherit model bias, prompt sensitivity, and blind spots.
6. **CLIP misses relation, count, and temporal errors.** Gates are strongest for object/scene grounding and weaker for subtle relational lies.
7. **Hallucination drop needs an anchor.** The fully no-anchor hallucination gate fails. Generated-anchor evidence is promising but not yet the headline.
8. **TRIBE is indirect.** It is trained on naturalistic brain responses, not BLV user outcomes. Its supported role is triage and routing hypotheses.

9. **Short clips.** Long-form narrative memory, timing slots, user preferences, and cognitive load remain future work.
10. **Human review remains required.** SceneTwin should route and assist human review; it should not replace professional describers.

Reproducibility

Primary artifacts:

- `cursor/output/external_ensemble_eval.csv` — 60-clip primary scores.
- `output/scenetwin_timing_20clip/ensemble/adqa_clip_ensemble_scores.csv` — 18-clip pilot scores.
- `cursor/output/corrected_ladder_robustness.json` — robustness checks.
- `cursor/research/recompute_corrected_ladder.py` — regenerates all corrected-ladder headline numbers (p, ordering, pairwise, permutation, bootstrap, per-category) from the per-tier CSVs; stdlib-only, deterministic.
- `cursor/output/gate_summary.json` — hallucination grounding-drop gate.
- `cursor/output/ref_subst/reference_substitution.json` — anchor substitution.
- `cursor/output/wrong_content_global_gate.json` — wrong-content gate.
- `cursor/output/gate_shipbest_selective.json` — selective ship-best.
- `cursor/research/output/vlm_as_judge_leaderboard.csv` — VLM judge comparison.
- `cursor/research/output/parallel_research/cheap_baselines/auc_summary.csv` — TRIBE cheap-baseline gauntlet.
- `cursor/research/output/parallel_research/access_surface/` — access-surface routing tables.

Supporting reports:

- `output/reports/scenetwin-paper-evidence-check.md`
- `output/reports/scenetwin-citation-map.md`
- `output/reports/parallel-research-cheap-baseline-gauntlet.md`
- `output/reports/parallel-research-access-surface-triage.md`
- `output/reports/parallel-research-synthesis.md`
- `output/reports/tribe-claims-audit.md`

Conclusion

SceneTwin is a human-reference-free audit framework for audio description. On a corrected three-tier ladder, it ranks AD candidates strongly on a 60-clip primary set, corroborated by the original 18-clip pilot. But its central contribution is not overwhelming leaderboard superiority. The central contribution is deployable audit infrastructure: competitive ranking without a professional reference AD, safety gates for hallucination and wrong-content failures, and brain-grounded triage for deciding where human review should go first. This is the layer AI-assisted AD workflows need before generated descriptions can be shipped responsibly at scale.

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